

Worktual Agentic Flow Guide

Complete reference: agents, tools, responsibilities, and execution paths
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1. Core Principle

Worktual uses a **two-layer architecture**:

Layer	Role
Gemini (LLM)	Routing, planning, coding, review, conversation
Python (backend)	Identity, tool filtering, validation, staging, QA, commit, rollback

Entry point: User message → `backend/api/generation.py` → `backend/agents/orchestration/runner.py`

Source of truth: `backend/AGENTIC_EXECUTION_FLOW.md` , `backend/agents/agent_tool_catalog.py` ,
`backend/agents/runtime_agents/registry.py`

2. End-to-End Flow

```
User message
  → generation API
  → Intent Router Agent
  → Intent? (greeting | simple_code | needs_confirmation | website_generation | website_update
| ...)
  → [optional] Requirement Confirmation Agent (confirm / revise / cancel)
  → Execution pipeline:
      website_generation → Greenfield Generation Engine → Streaming File Agent / Parallel
Workers
      website_update      → ScopeEngine + Streaming File Agent
      simple_code          → Simple Code Writer Agent
  → Commit (WRITE_PROJECT_FILES)
  → Memory Agent (persist summary)
```

3. Routing Intents

Classified by **Intent Router Agent** (`backend/agents/orchestration/routing.py`).

Intent	Meaning	Active Agent
greeting	Hello / small talk	Greeting Handler Agent
needs_more_detail	Vague build request	Conversation Agent
project_info	Explain / summarize project	Conversation Agent (read-only)
simple_code	Standalone algorithm/script	Simple Code Writer Agent
needs_confirmation	Brief ready, waiting	Requirement Confirmation Agent
website_generation	Build new website/app	Greenfield Generation Engine
website_update	Change existing project	Streaming File Agent + ScopeEngine

4. Orchestration Agents (Pre-Execution)

4.1 Intent Router Agent

Tools	<code>route_generation_action</code>
Responsibility	Classify every turn → intent + next tool (strict JSON)
File	<code>backend/agents/orchestration/routing.py</code>

4.2 Greeting Handler Agent

Tools	<code>handle_greeting</code>
Responsibility	Warm reply; collect business type, pages, style

4.3 Requirement Confirmation Agent

Tools	<code>confirm_execution_brief</code>
Responsibility	Build execution brief; block writes until user confirms
Decisions	confirm revise cancel new_request unclear
Files	<code>backend/agents/requirement_confirmation/</code>

4.4 Simple Code Writer Agent

Tools	<code>generate_simple_code_file</code>
Responsibility	Python/Java/etc. single files — no Vite/React scaffold

4.5 Streaming File Agent (Primary Coder)

Tools	<code>read_file</code> , <code>write_file</code> , <code>str_replace</code> , <code>list_files</code> , <code>search_codebase</code>
Responsibility	Live tool-calling loop — read project, edit files, finish with summary
Used for	Website updates (default), greenfield builds, parallel worker subprocesses
File	<code>backend/agents/streaming/file_agent.py</code>

4.6 Greenfield Generation Engine

Responsibility	Inject scaffold → plan file blueprint → generate pages/components/routes → validate → auto-resume if incomplete
File	<code>backend/agents/generation_engine/greenfield_runner.py</code>

4.7 Parallel Stream Orchestrator (Legacy)

When	ENABLE_LEGACY_PARALLEL_UPDATES=true + parallel flags
File	backend/agents/streaming/parallel_orchestrator.py

4.8 Parallel File Workers

When	Rich CRM / module-heavy greenfield (ENABLE_GREENFIELD_PARALLEL_WORKERS)
File	backend/agents/streaming/parallel_file_workers.py

4.9 ScopeEngine (Unified Update Authority)

Responsibility	Sole file-routing for updates: memory + code index + LLM → candidate files, enrichment, interaction metadata
File	backend/agents/update_engine/scope_engine.py

4.10 Specialist Agents (Optional)

Agent	Role
Content Specialist Agent	Copy, sections, tone JSON plan
Layout Specialist Agent	Pages, components, layout plan
Catalog Specialist Agent	Entities, fields, catalog plan

Skipped for website_generation / website_update in default config.

5. Full Runtime Loop Agents (16)

Run when LangGraph / legacy runtime is enabled and streaming fast path is **not** selected. Supervisor picks actions from `ACTION_REGISTRY`.

Agent	Action(s)	Backend Tool(s)	Responsibility
Supervisor Agent	DONE	—	Pick next legal action; stop when proof complete
Memory Agent	READ/LOAD/PERSIST	Same	Load context; persist summary after commit
Universal Error Handling Agent	RUN_ERROR_HANDLING_AGENT	—	Triage build/runtime/API errors
Update Analysis Agent	RUN_UPDATE_ANALYST	—	Choose update mode + candidate files
Scoped Update Agent	RUN_SCOPED_UPDATE_AGENT	APPLY_PATCH	Patch only approved paths
Prompt Analyst Agent	RUN_PROMPT_ANALYST	—	Structured brief from prompt + files
Planner Agent	RUN_PLANNER	—	Section/layout/implementation plan
Agent Registry Agent	RUN_DYNAMIC_*	Read only	Plan/run dynamic specialists
UX Review Agent	RUN_UX_REVIEW_AGENT	—	Conversion, layout, workflow risks
Accessibility Agent	RUN_ACCESSIBILITY_AGENT	—	Contrast, semantics, keyboard
Code Agent	RUN_CODE_AGENT	—	Generate candidate artifact JSON
Code Generator Agent	RUN_DYNAMIC_PATCH_INTEGRATOR	—	Merge dynamic-agent patches
Materialize Agent	MATERIALIZE_CANDIDATE_FILES	WRITE_PROJECT_FILES	Stage files before preview
Repair Agent	RUN_REPAIR_AGENT	—	Fix after validation/QA failure
Validation Agent	VALIDATE_PROJECT_ARTIFACT	VALIDATE_PROJECT_ARTIFACT	Contract/shape/path checks
Preview Agent	BUILD_STAGED_PROJECT_PREVIEW	BUILD_STAGED_PROJECT_PREVIEW	Staged Vite build
Visual QA Agent	RUN_PREVIEW_VISUAL_QA	RUN_PREVIEW_VISUAL_QA	Browser/preview integrity

Commit Agent	WRITE_PROJECT_FILES	WRITE_PROJECT_FILES	Persist after all gates pass
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Default today: Full loop is off for website intents when streaming is enabled.

6. Dynamic Agent Registry

Core (7): Intent Analyzer, Requirement Analyst, Task Decomposer, Agent Registry, Workflow Planner, Supervisor, Memory

Reusable specialists (8): Domain Research, UX/Layout, Content, Component/UI, Code Generator, Validation, Preview QA, Repair

Tool policy: Dynamic agents may only use `READ_PROJECT_FILES` and `LOAD_PROJECT_MEMORY` — no writes.

When: `ENABLE_FULL_DYNAMIC_GENERATION=true`

7. All Tools

7.1 Streaming Tools (Gemini — Fast Path)

Tool	What it does
<code>read_file</code>	Read project file content
<code>list_files</code>	List directory / project files
<code>search_codebase</code>	Hybrid code index search
<code>write_file</code>	Create or replace entire file
<code>str_replace</code>	Surgical edit on existing file

7.2 Backend Executable Tools (Runtime Loop)

Tool	Purpose
READ_PROJECT_FILES	Load all project files
LOAD_PROJECT_MEMORY	Load episodic/session memory
PERSIST_PROJECT_MEMORY	Save run summary to memory
WRITE_PROJECT_FILES	Commit staged files (after gates)
VALIDATE_PROJECT_ARTIFACT	Schema + syntax validation
BUILD_STAGED_PROJECT_PREVIEW	Vite preview build
RUN_PREVIEW_VISUAL_QA	Screenshot/visual QA
STR_REPLACE, APPLY_PATCH	Edits
SEARCH_CODEBASE, READ_FILE, LIST_DIR	Platform file ops
RUN_TERMINAL, GIT_*, RUN_TESTS, RUN_LINT	Dev execution

7.3 Orchestration Pseudo-Tools

Tool	Maps to
<code>route_generation_action</code>	Intent Router
<code>handle_greeting</code>	Greeting Handler
<code>confirm_execution_brief</code>	Confirmation Agent
<code>generate_simple_code_file</code>	Simple Code Writer
<code>analyze_prompt</code> / <code>analyze_update_request</code>	Legacy artifact pipeline

8. Default Production Paths

Website Generation

Intent Router → Confirmation (if enabled) → Greenfield Generation Engine

- plan_greenfield_parallel_tasks
- Streaming File Agent + blueprint OR Parallel File Workers (rich CRM)
- Minimum deliverable check → auto-resume if scaffold-only
- Syntax guard → Vite build → Visual QA → WRITE_PROJECT_FILES

Website Update

Intent Router → [optional Confirmation] → Streaming File Agent

- ScopeEngine (memory + code index + LLM)
- Scope enrichment (handlers, styles, interaction wiring)
- str_replace / write_file on scoped files
- Commit pipeline + gates

Simple Code

Intent Router → Simple Code Writer Agent → single artifact → validate

Conversation

Intent Router → Conversation Agent → no file writes

9. Feature Flags

Flag	Effect
ENABLE_UNIFIED_UPDATE_ENGINE	ScopeEngine for updates (default on)
ENABLE_LEGACY_PARALLEL_UPDATES	Old parallel/LangGraph paths (default off)
ENABLE_STREAMING_FILE_AGENT	Fast streaming coder (default on)
ENABLE_GREENFIELD_PARALLEL_WORKERS	Parallel page workers for rich CRM (default on)
REQUIRE_PLAN_CONFIRMATION	Confirmation Agent before generation
ENABLE_LANGGRAPH_WEBSITE_RUNTIME	Full LangGraph supervisor loop
ENABLE_FULL_DYNAMIC_GENERATION	Dynamic specialist registry
ENABLE_PATCH_APPROVAL	Pause before commit for user approval
ENABLE_CODE_INDEX	search_codebase + scope retrieval

10. Adaptive Routing

Route	Typical Execution
tiny_chat	No model call
small_code	Minimal standalone code
targeted_update	Scoped streaming patches
feature_update	Multi-file staged patches
large_project	Parallel workers (if legacy enabled)
full_generation	Greenfield parallel or blueprint agent

11. CRM Use-Case Quick Reference

Need	Agent(s)
Build CRM with auth → onboarding → dashboard	Intent Router → Confirmation → Greenfield Engine → Streaming/Parallel
Fix cart button	Intent Router → ScopeEngine → Streaming File Agent + interaction enrichment
Add FastAPI backend	Greenfield planner adds backend/*.py → Streaming File Agent
Change one color	ScopeEngine → targeted patch (may skip confirmation)
Write Python prime checker	Simple Code Writer Agent

12. Key Source Files

Topic	File
Agent names list	docs/available_agent_names.md
Agent ↔ tool metadata	backend/agents/agent_tool_catalog.py
Runtime action map	backend/agents/runtime_agents/registry.py
Orchestrator decisions	backend/agents/orchestration/runner.py
Greenfield planning	backend/agents/streaming/task_planner.py
Update scope	backend/agents/update_engine/scope_engine.py
Executable tools	backend/agentic/tools/registry.py
Execution contract	backend/AGENTIC_EXECUTION_FLOW.md

Worktual Agentic Flow Guide — internal architecture reference. For the latest behavior, verify against runtime_config.py and agent_tool_catalog.py in the repository.